

Course Content

	Topics
1	Computer Hardware Decomposition Interconnection of processor, memory, data storage, power supplies and peripherals. Clock and reset circuitry. Input devices such as touch screens, mice, and keyboards and output devices.
2	Data Representation, Memory Allocation and Access Signed and unsigned number representation. Little Endian vs Big Endian. Character encoding (DEC 6-bit vs ASCII vs Unicode). Arrays, string (C and Pascal), structure and pointer representation. Data alignment and data padding.
3	Central Processing Unit Concept of a stored program. Allocation of memory to code and data segments in Von Neumann architecture and dual memory spaces in Harvard architecture with concurrent access. Programmer's functional models of processors. Status register and flags. Control unit. Fetch-decode-execute cycle.
4	Assembly Programming and Instruction Set Architecture Assembly-level programming. Addressing modes. Use of Program Counter for instruction and data access. Basic arithmetic and logical operations. Program control instructions. Implementing position-independent code. Supervisor vs User modes. Privileged instructions. Instruction set architectures (ISA) and paradigms. Classifications based on accumulators, registers, load-store and memory-to-memory transfers and operations. Differentiating characteristics of 8-bit to 64-bit ISAs. Fixed-length vs variable-length instructions. Mnemonics, instruction encoding and orthogonality.
5	High-level Software to Low-level Instructions Mapping of high-level language constructs for control flow and data structures to machine-level operations. Including 'if', 'switch', 'for' and 'while' with pre- and post- test conditions. Using examples of code produced by a 'C' compiler. Compiler optimisation techniques (e.g. jump tables, loop unrolling) for execution speed or code size. Modular programming, mechanisms for subroutine/function linkage, including return address, parameter/result passing and local data. Parameter passing using register, memory and stacks. Transparent subroutines. Implementing local variables with stack frame and frame pointer.
6	Signal Chain Sub-System Interfacing to the Real World - Analog to digital conversion - Digital sampling process - Interpreting digital logic levels - Digital to Analog conversion Serial versus parallel data transfer. Asynchronous versus synchronous digital data transfer. Digital Serial Interface Standards e.g. UART, SPI. Sources of interference to data transmission e.g. Signal Skews, Cross Talk. Programmed I/O. Interrupts – flow mechanism and interrupt service routines. Direct Memory Access – configuration, flow and notification mechanism

7	Memory Sub-System Operating principles, design and interface consideration - Volatile and non-volatile Memory - Static RAM and Dynamic RAM - Flash and EEPROM - Magnetic HDD and Solid State Drive Cache Memory Management - Operating principles - Cache Mapping Scheme (Direct Mapped) Virtual Memory Management - Physical and Virtual Memory Addressing Space - Virtual Memory Management: Paging Scheme Integration of Cache and Virtual Memory
8	Communication and User Interface Sub-System Wired, e.g. USB, SATA, HDMI Wireless, e.g. WIFI, Bluetooth Input interface, e.g. Key Board, Mouse, Capacitive Touch, Camera, Microphone. Output interface, e.g. Display and Speakers
9	Computer Arithmetic Positional numbering systems. Fixed Point Number System - 2's complement system. - Fixed-point arithmetic. - Carry vs. Overflow. Floating point number system - IEEE 754 standard. - Floating-point arithmetic. Fixed Point vs Floating Point Number system Implementation consideration - Effects of the four operators (+, -, *, /) - Effects of rounding. - Handling of numbers with different magnitude - Maximising precision during computation
10	Performance Optimization Processor Pipeline Architecture - Basic operating concept - Pipeline conflicts and management - Performance computation